

ECSE-4040

Digital Electronics

Lecture 20

Bipolar Digital Circuits

Profs. Mona Hella and T. Paul Chow
Rensselaer Polytechnic Institute
Troy, NY 12180-3590

References: Appendix B: HJS

Chapters 5 and 17, Neamen

<https://lms.rpi.edu>

Lecture Outline

- Modified Low Power ECL
- Series Gating
- BiCMOS Technology

High Speed Logic Families

Emitter-Coupled Logic (ECL)

- Bipolar
- III-V FET
- BiCMOS



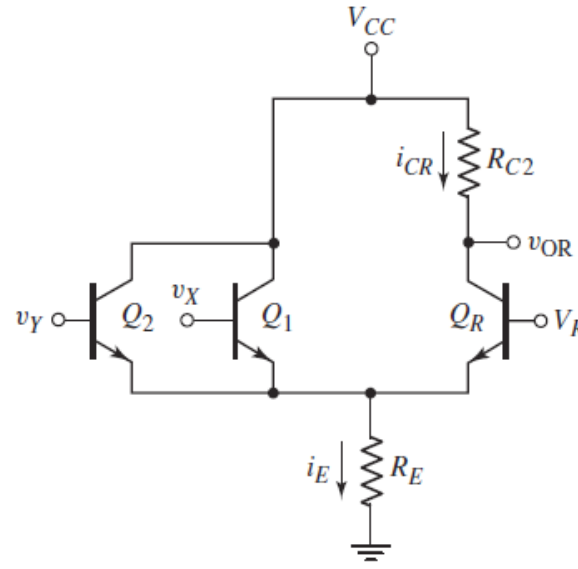
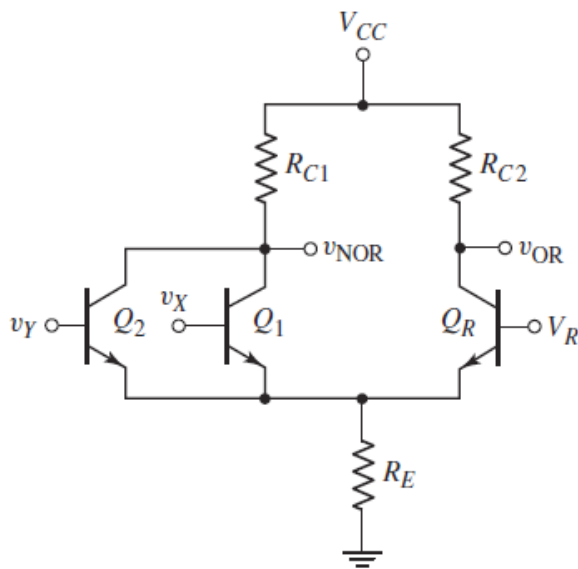
Current-Mode Logic (CML)

- ☐ Bipolar
- ☐ MOS
- ☐ BiCMOS

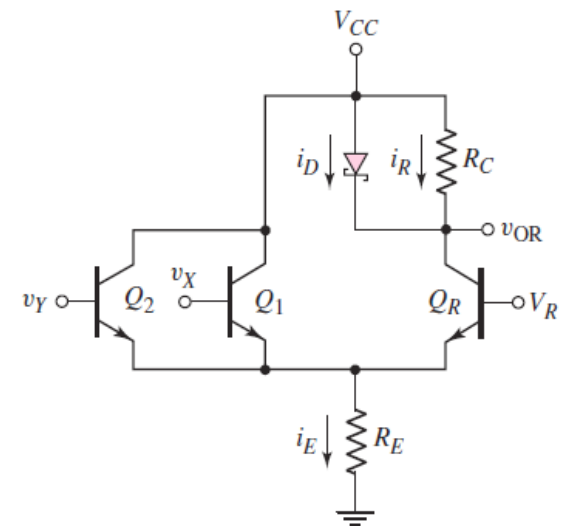


**Some high-speed logic families use differential topologies

Low Power ECL



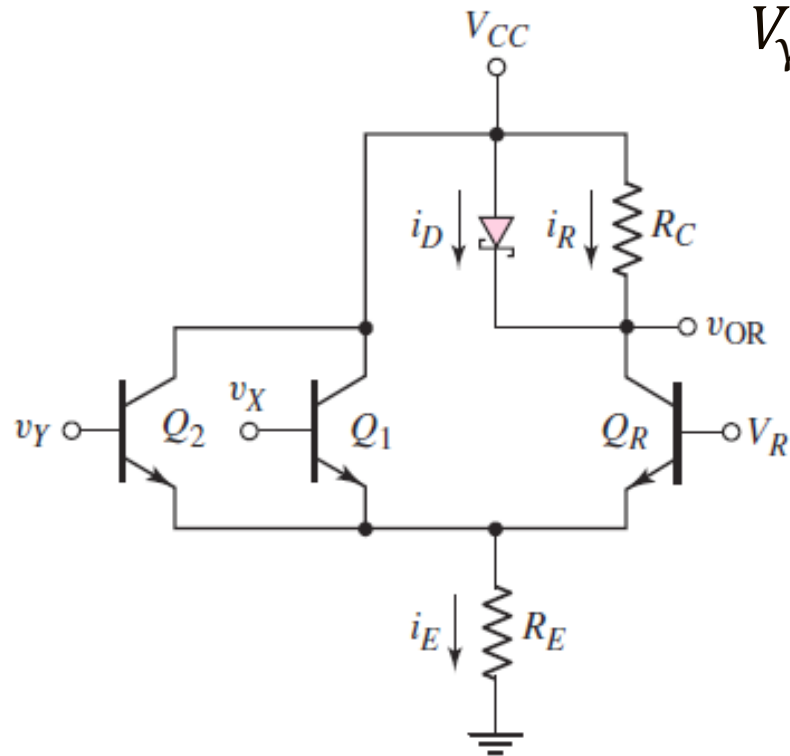
Logic '0' depends on R_{C2}



Logic '0' set by V_{CC} and turn-on voltage of diode

- Make the output voltage states compatible with the input voltages, without the emitter-follower output stages
- If complementary outputs are not required, eliminate resistor R_{C1} (will not affect the power consumption).
- Eliminate the dependence on R_E and R_C

Low Power ECL (OR Gate)



V_Y turn on voltage of the Schottky diode

$$V_x, V_y < V_R \quad Q_1, Q_2 \text{ OFF}$$

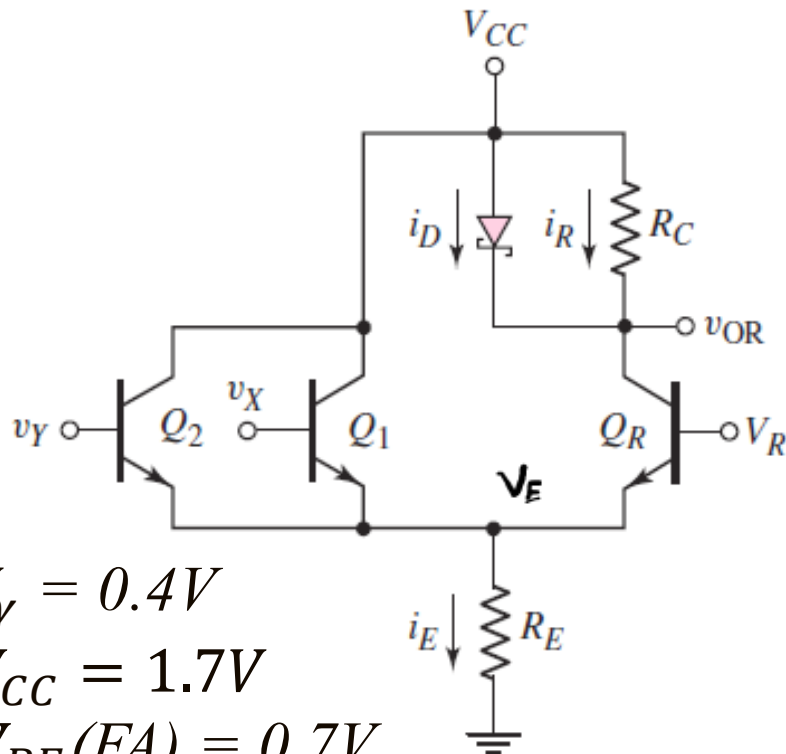
$$V_{OL} = V_{CC} - V_Y$$

$$V_x, V_y > V_R \quad Q_1, Q_2 \text{ ON}$$

$$V_{OH} = V_{CC}$$

$$V_R = V_{CC} - \frac{V_Y}{2}$$

Low Power ECL (OR Gate)



$$V_Y = 0.4V$$

$$V_{CC} = 1.7V$$

$$V_{BE}(FA) = 0.7V$$

$$V_{OH} = V_{CC} = 1.7V$$

$$V_{OL} = V_{CC} - V_Y = 1.3V$$

$$V_R = V_{CC} - \frac{V_Y}{2} = 1.5V$$

$$V_x, V_y < V_R \quad Q_1, Q_2 OFF$$

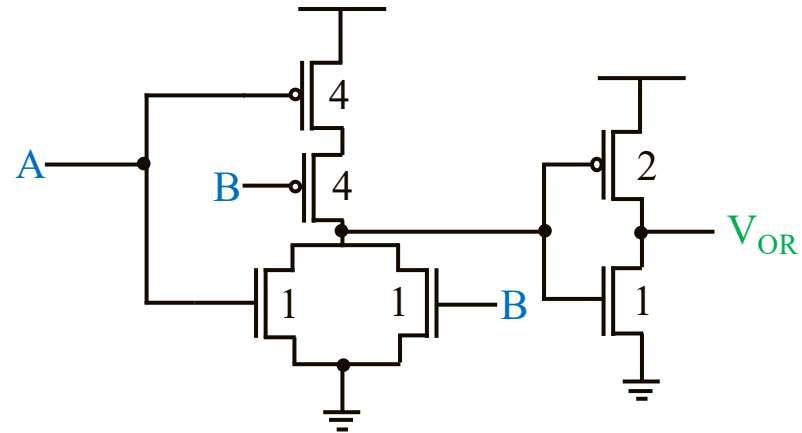
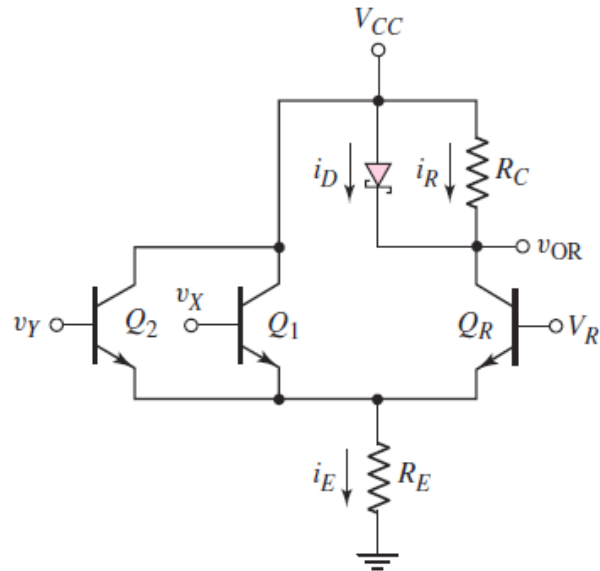
$$V_E = V_R - V_{BE}(FA)$$

$$V_E = 1.5 - 0.7 = 0.8V$$

$$V_x, V_y > V_R \quad Q_1, Q_2 ON$$

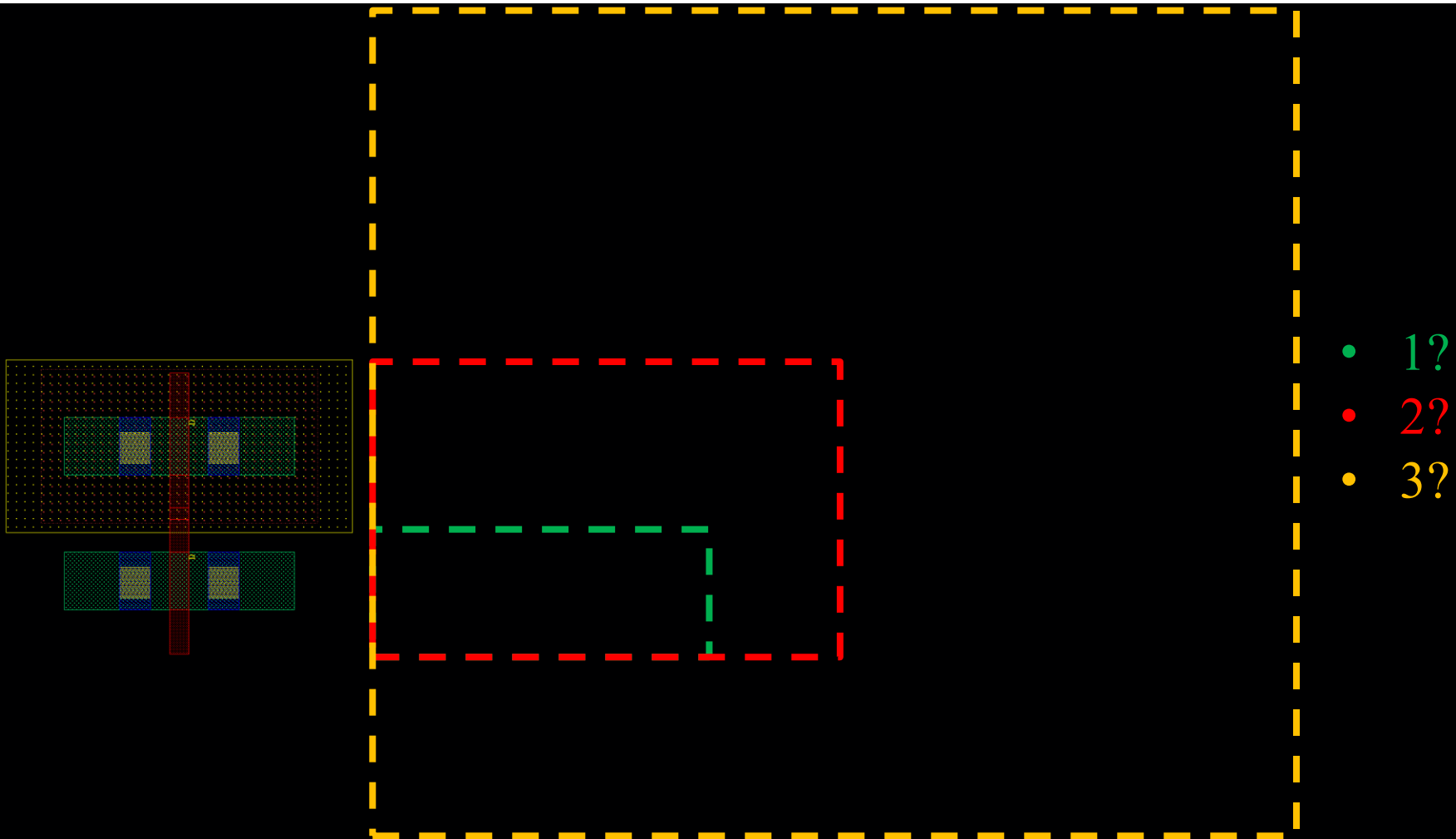
$$V_E = V_{OH} - V_{BE}(FA) = 1.0V$$

BJT circuits – how big is it?

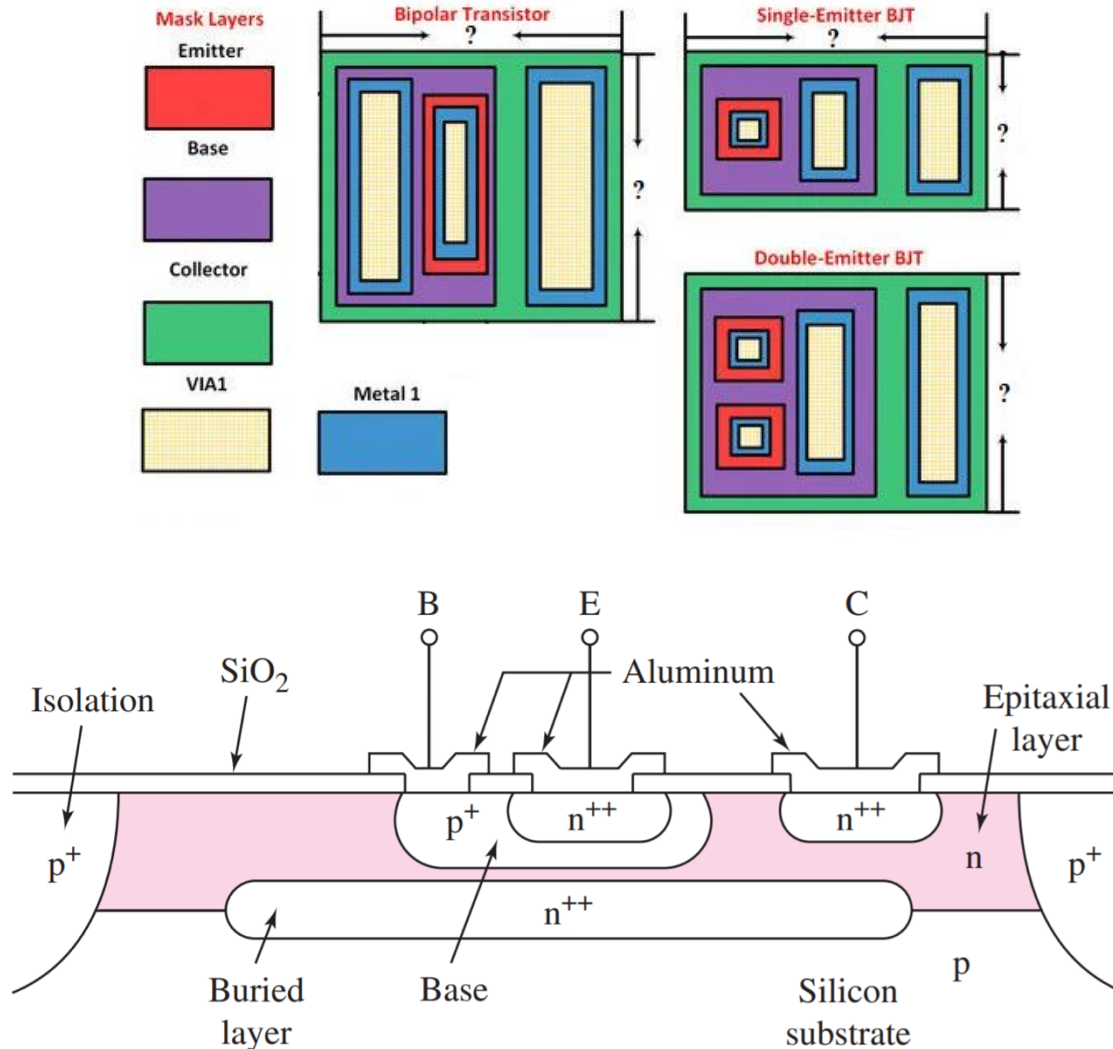


- ECL low-power OR is 3 transistors.
- CMOS OR is 6 transistors.
- This must mean that ECL circuits are smaller, right?

BJT – how big is it?

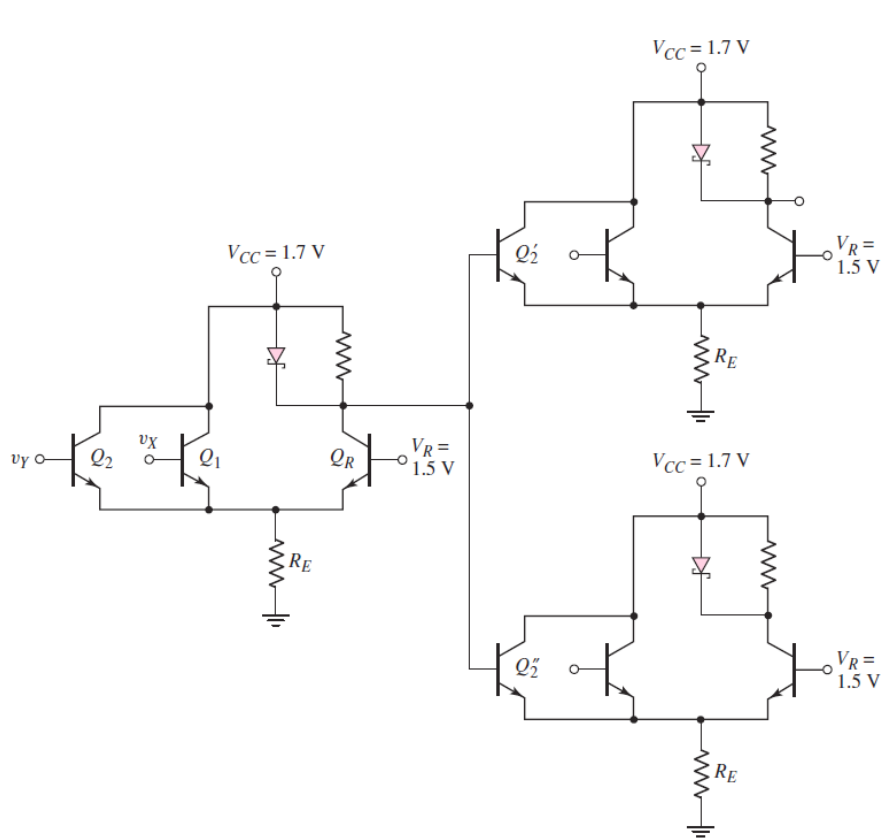


BJT – how big is it?

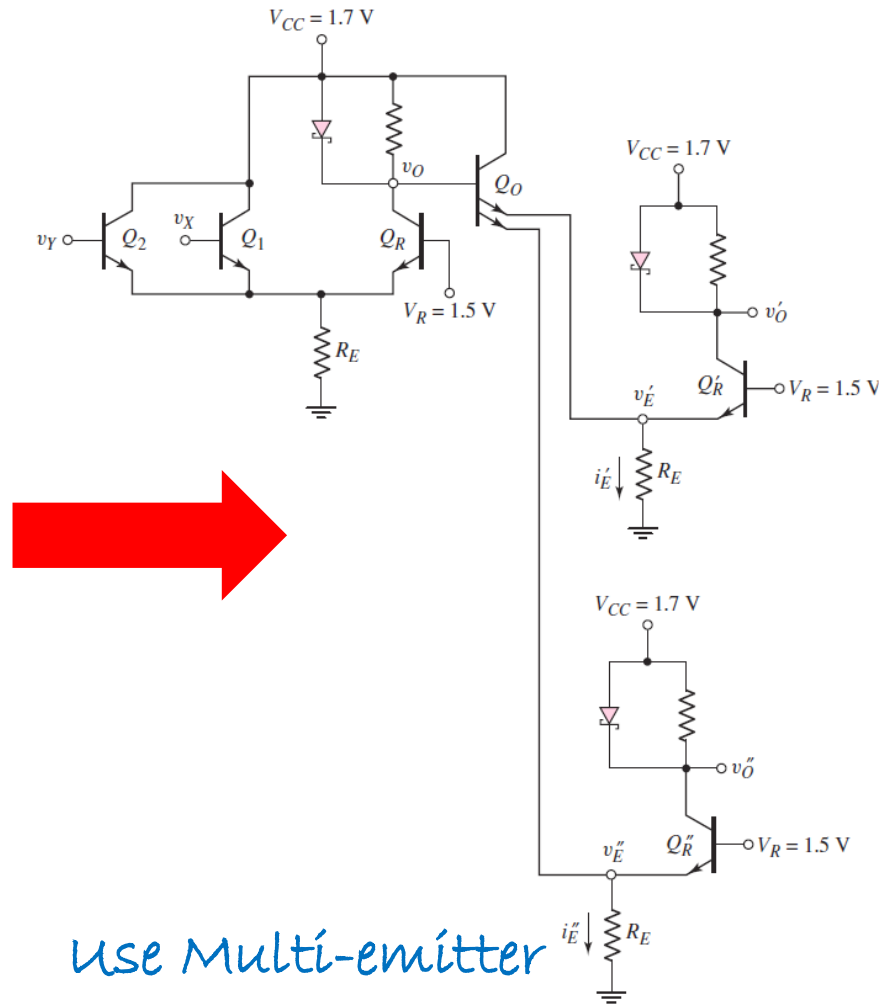


<https://www.mdpi.com/2079-9292/8/5/496/htm#>

Modified ECL for IC Implementations



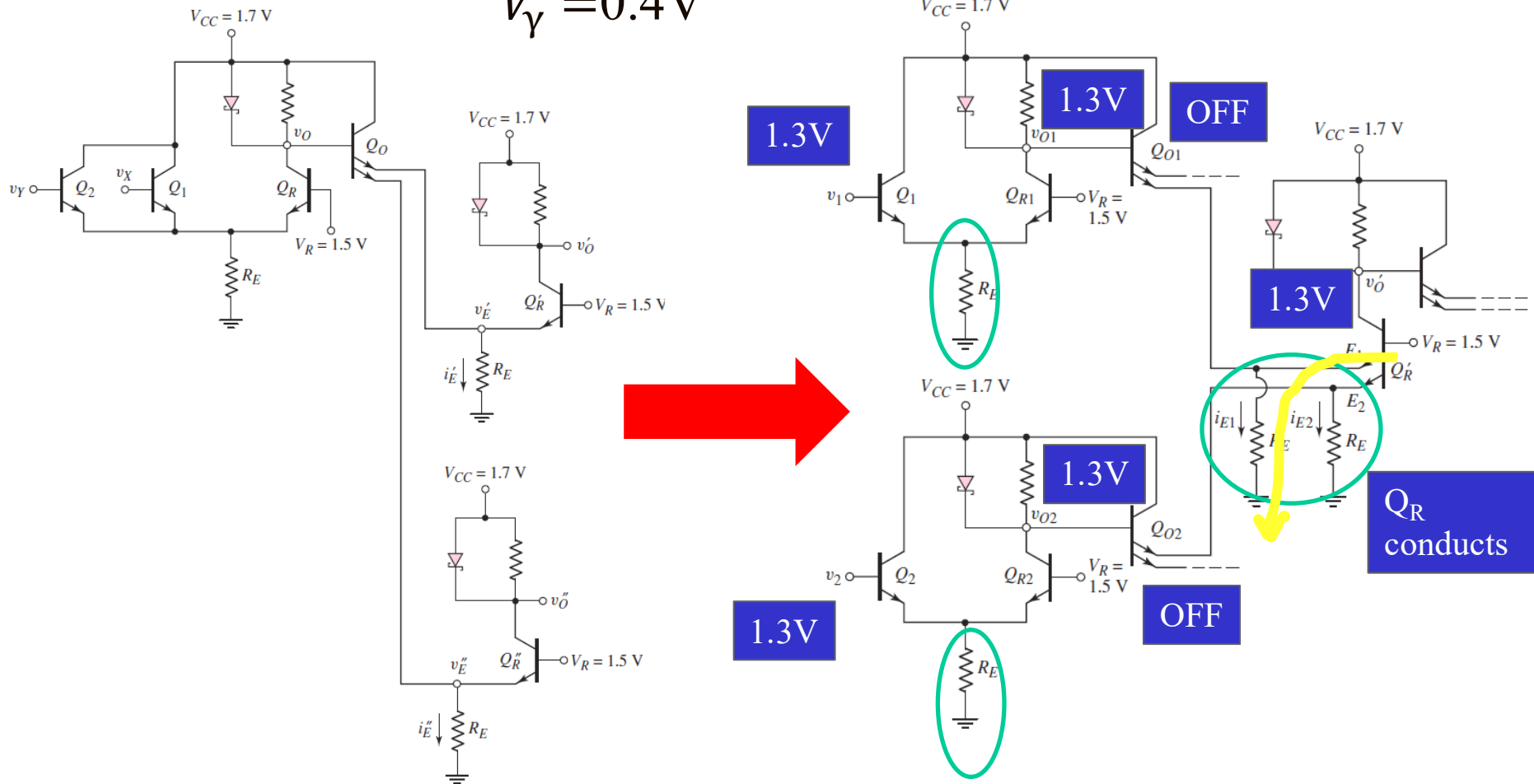
Collectors and bases of Q_2' and Q_2'' are the same



use Multi-emitter transistor

Modified ECL for IC Implementations

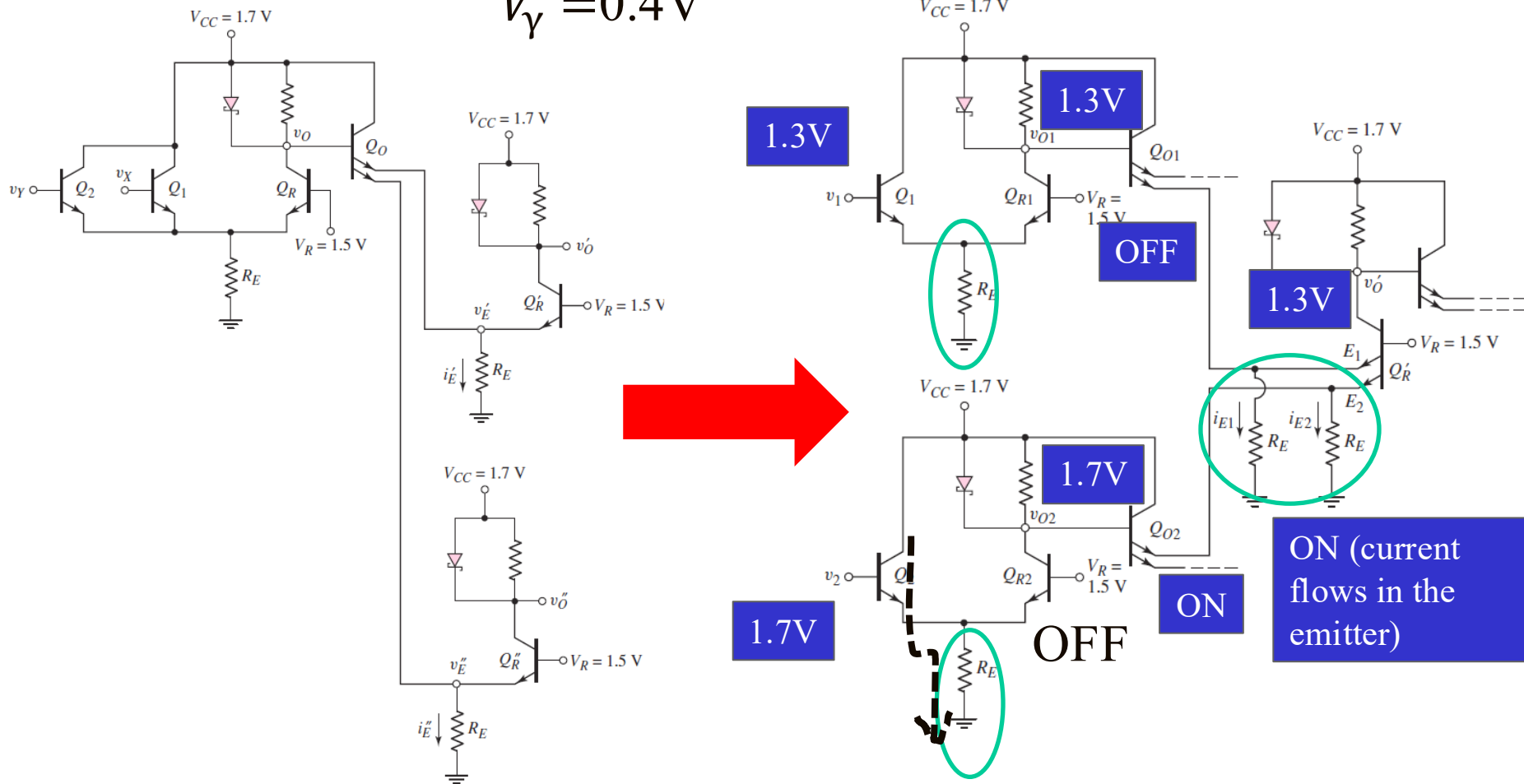
$$V_Y = 0.4V$$



$$V_{OH} = V_{CC} = 1.7V \quad V_{OL} = V_{CC} - V_Y = 1.3V$$

Modified ECL for IC Implementations

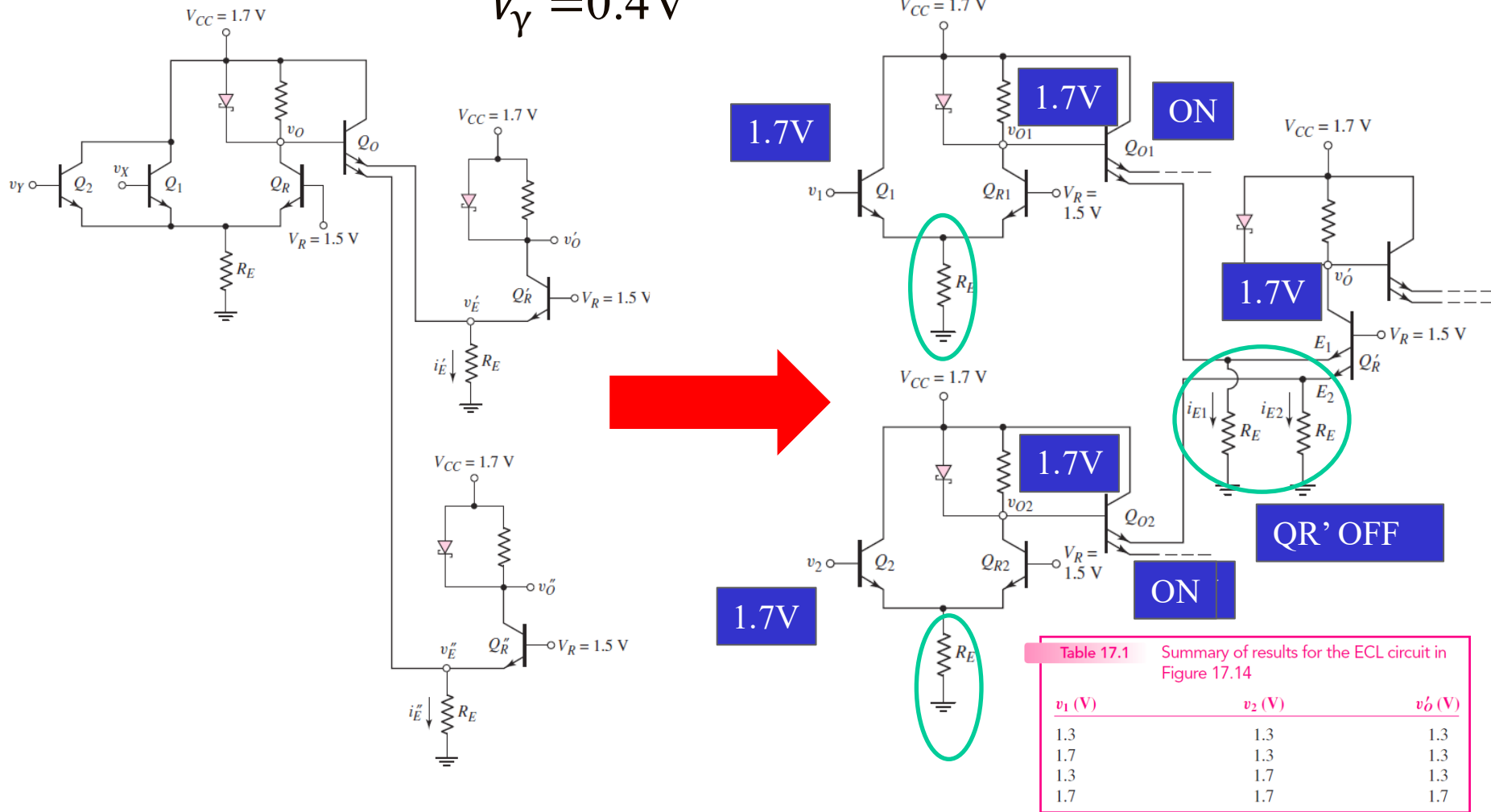
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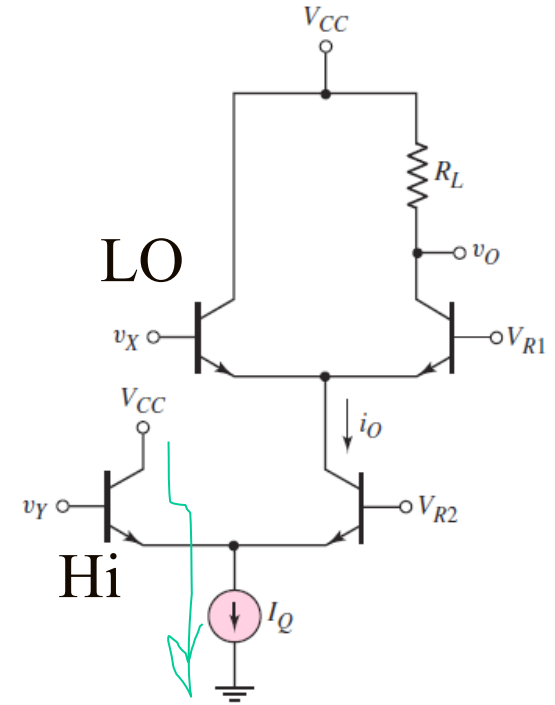
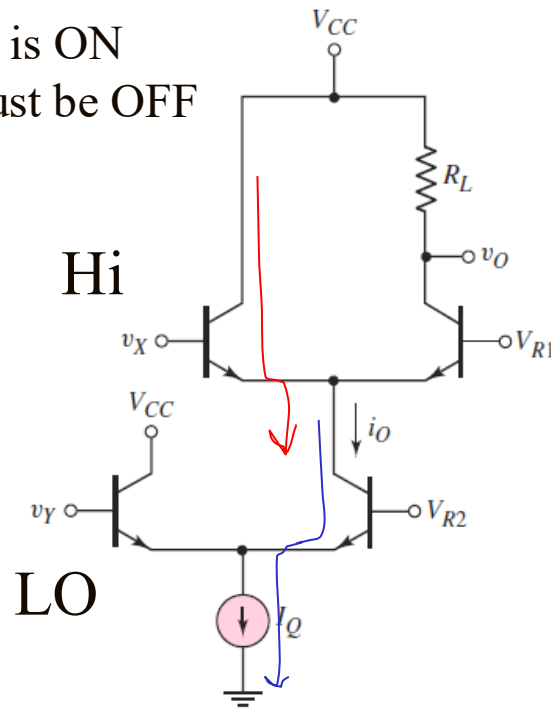
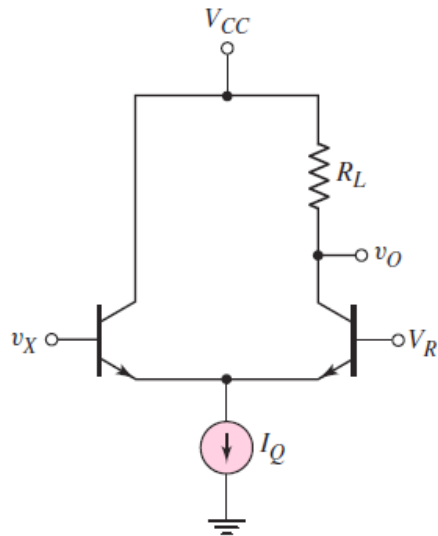
Modified ECL for IC Implementations

$$V_Y = 0.4\text{V}$$



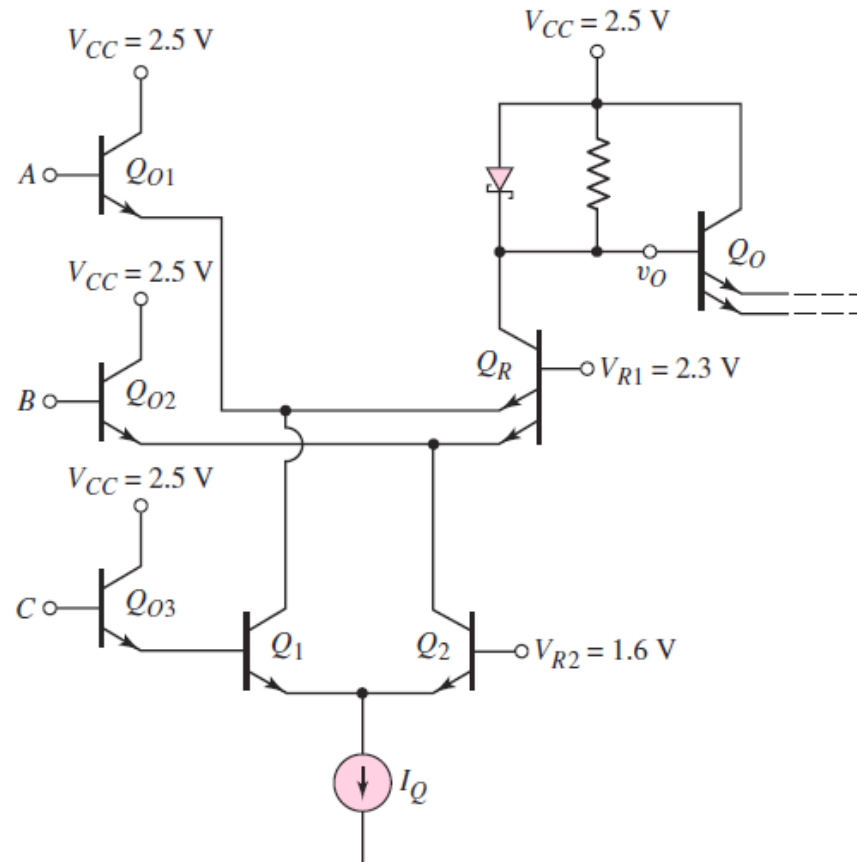
Series Gating-CASCODE

- Key concept:
 - I_E means branch is ON
 - Other branch must be OFF



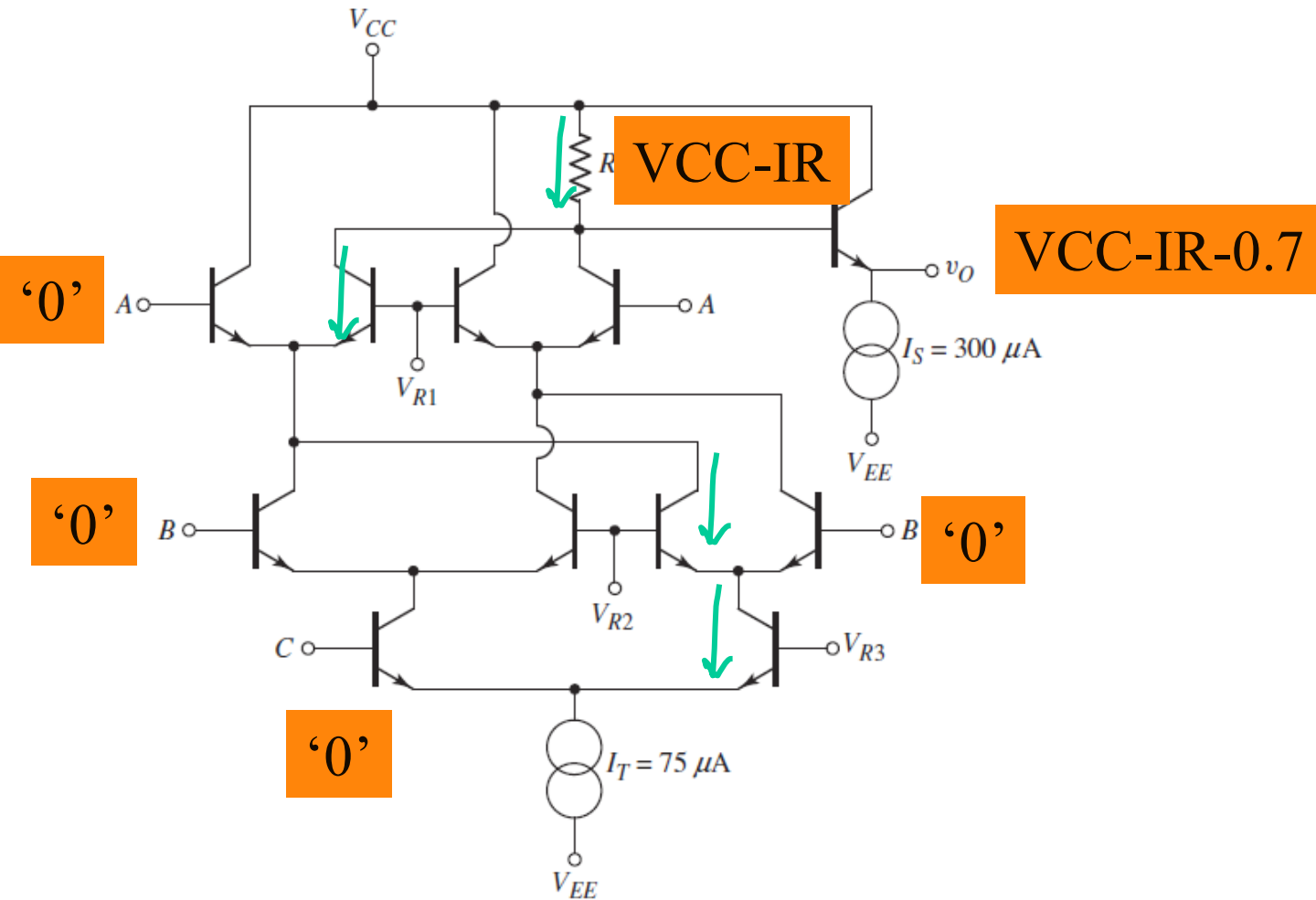
v_X	v_Y	v_O
LO	LO	$V_{CC} - I_Q R_L$
LO	Hi	V_{CC}
Hi	LO	V_{CC}
Hi	Hi	V_{CC}

Series Gating

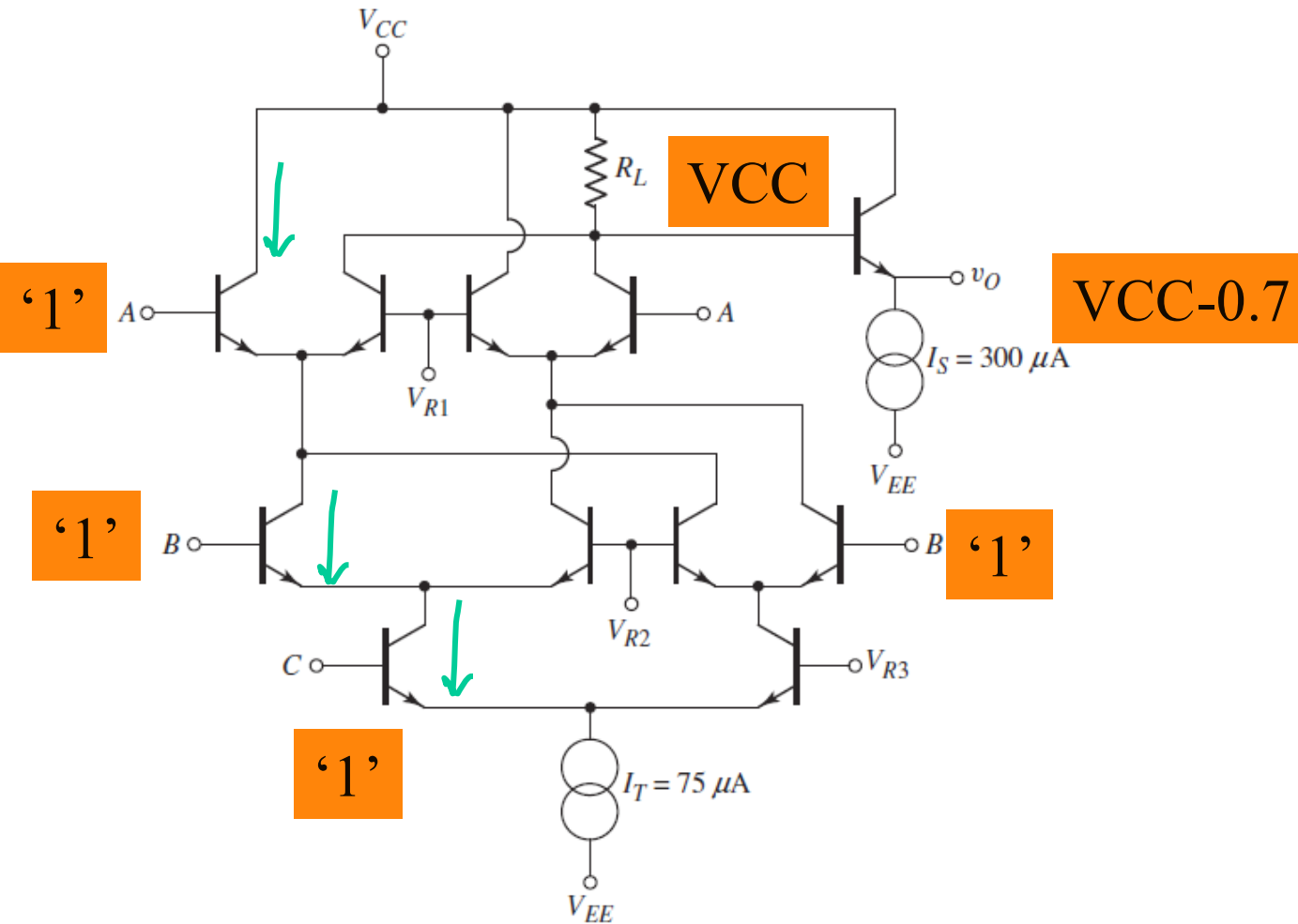


$(A \text{ AND } C) \text{ OR } (B \text{ AND } \bar{C})$

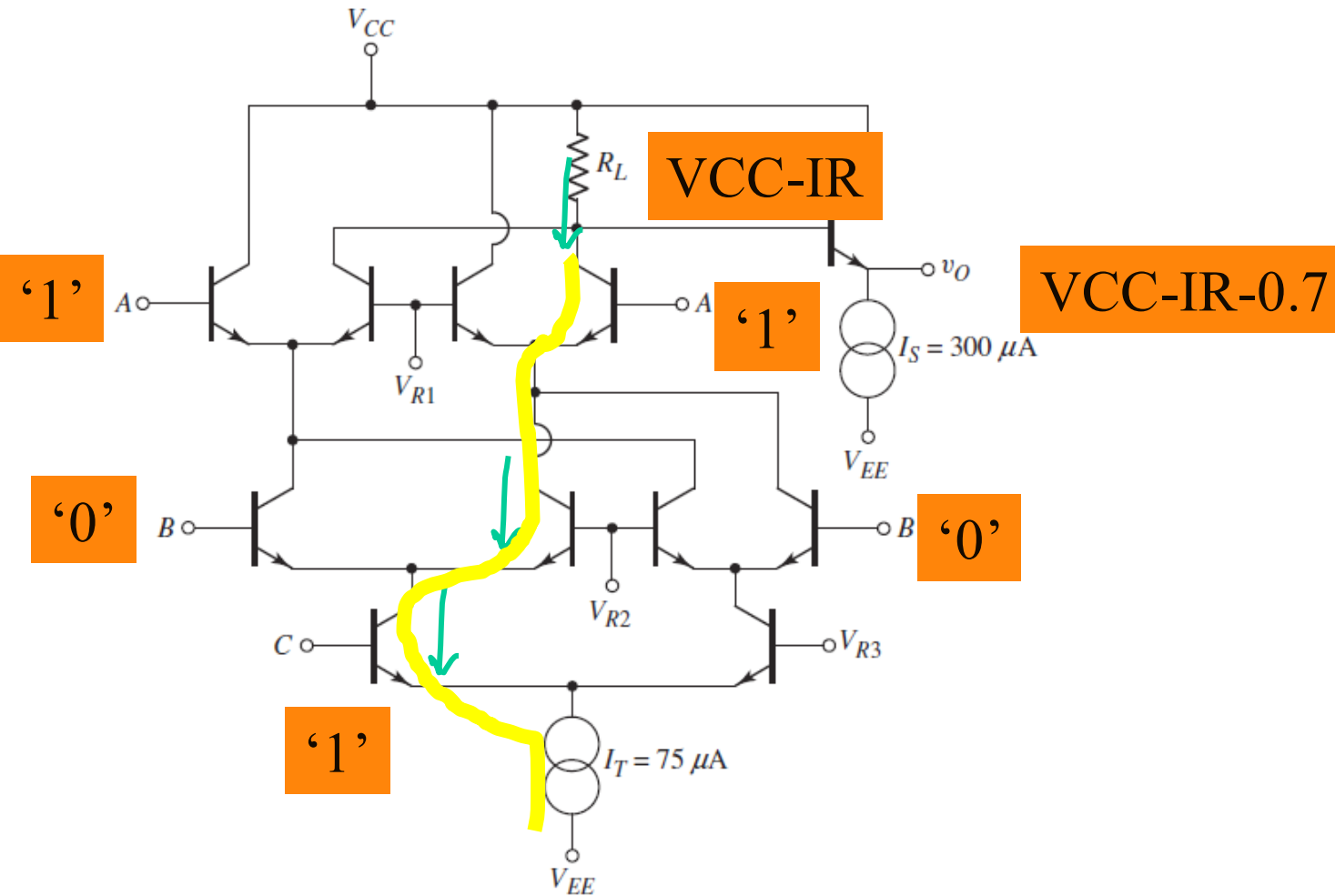
Another Example



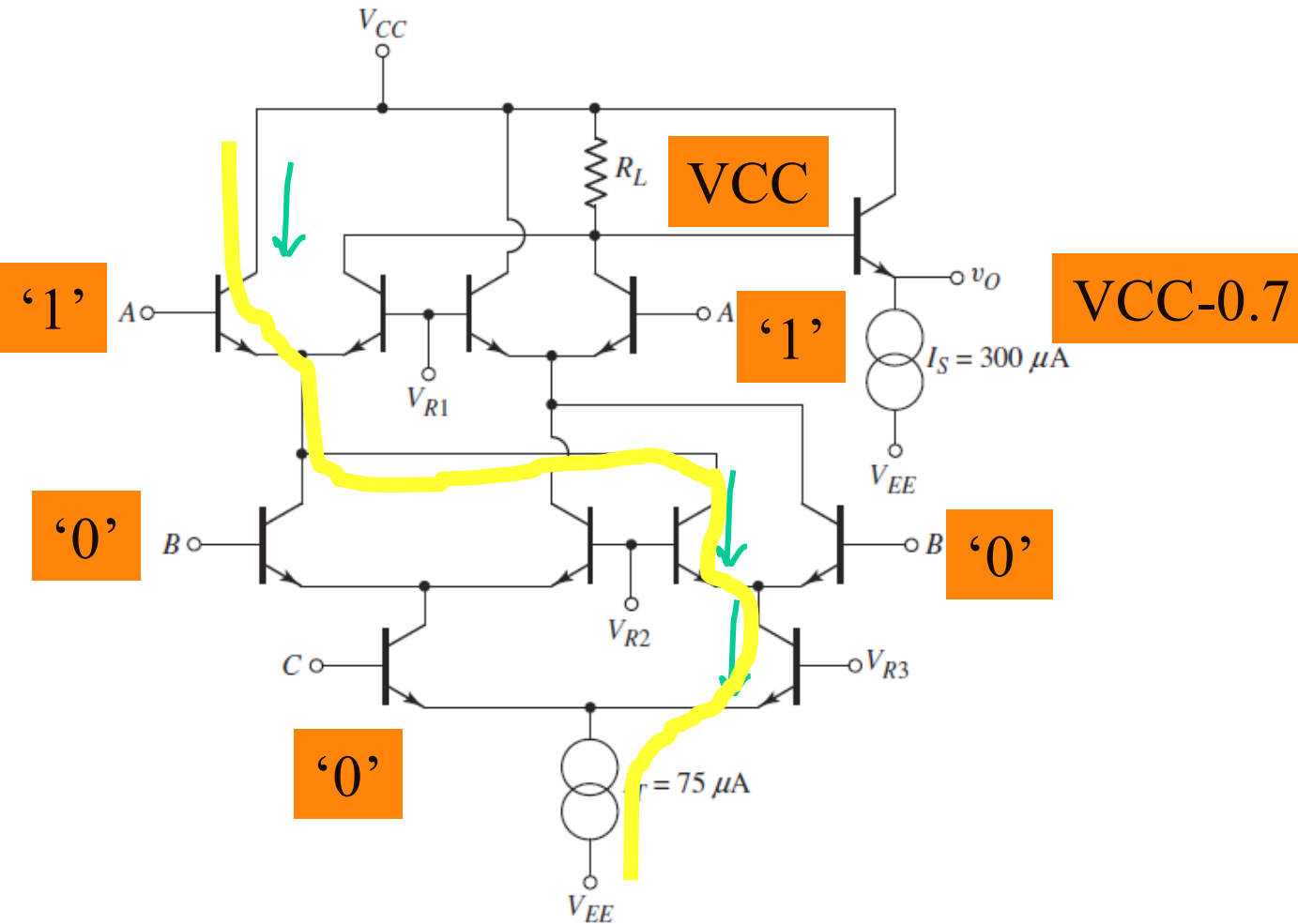
Another Example



Another Example

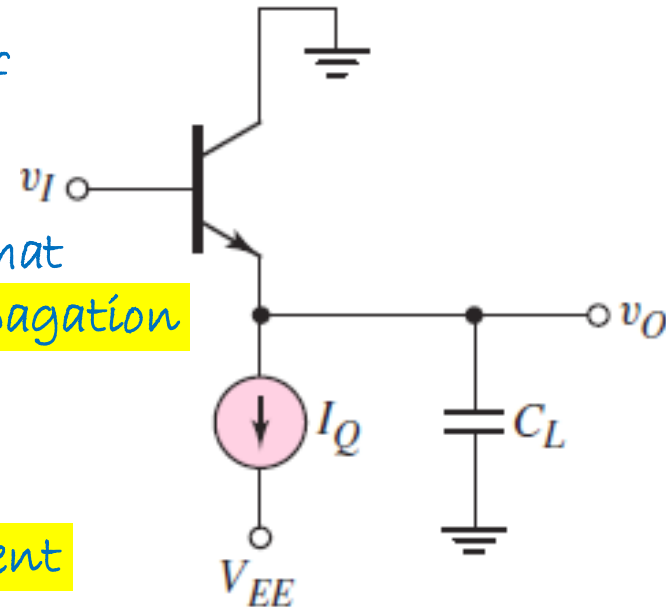


Another Example



Propagation Delay time

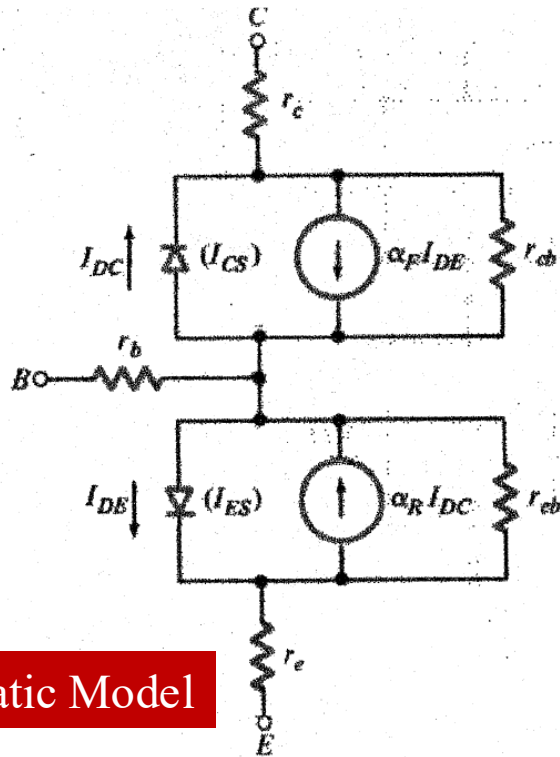
- ECL is the fastest bipolar logic technology.
- Bipolar technology can produce small, very fast transistors with cutoff frequencies in the range of 500GHz → 700GHz → 1THz.
- Logic gates that use these transistors are so fast that interconnect line delays tend to dominate the propagation delay times.
- Minimizing these interconnect delays involves minimizing the metal lengths and using sufficient current drive capability.
- Speed is derived from low-signal logic swings, non-saturating logic, and the ability to drive a load capacitance.
- Emitter follower? Why?



FA

Diffusion charge

Depletion charge



Static Model

$$Q_{BE} = \tau_F I_S (e^{V_{BE}/n_E V_T} - 1) + C_{jE0} \int_0^{V_{BE}} dV \left(1 - \frac{V}{\phi_E}\right)^{-m_E}$$

$$Q_{BC} = \tau_R I_S (e^{V_{BC}/n_C V_T} - 1) + C_{jC0} \int_0^{V_{BC}} dV \left(1 - \frac{V}{\phi_C}\right)^{-m_C}$$

$$V_{BE}(FA) = 0.7V$$

$$V_{BE}(sat) \approx 0.8V$$

$$\frac{e^{0.8/0.026}}{e^{0.7/0.026}} = \underline{47}$$

47x diffusion charge gain

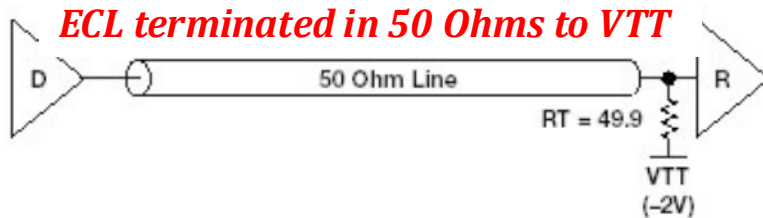
$$C = \frac{Q}{V} = \frac{47}{0.8/0.7} = \underline{41}$$

41x diffusion capacitance gain.

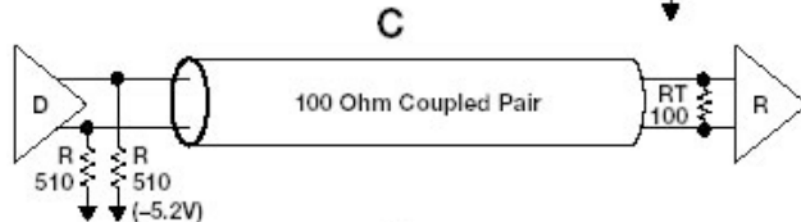
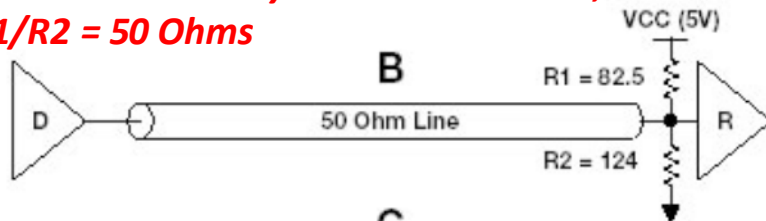
Summary

- ECL - Emitter Coupled Logic - is the oldest of high-speed logic families (early 1960s by Motorola -- (Motorola ECL) family).
- ECL has evolved into many enhanced families such as RSECL for Reduced Swing ECL (400mV output swing).
- ECL can be used single-endedly or differentially depending upon the application and noise margins needed.
- The drivers are low impedance open emitter outputs that generate a typical 700 to 800 mV output voltage.
- The output stage is operated in the active region, saturation is prevented, thus ECL's famous fast and balanced edge rates are obtained.
- The output is typically terminated with 50 Ohms to a termination rail that is two volts less than the more positive rail. ECL parts are commonly powered between ground and -5.2V.
- Due to the negative rail requirement and compatibility with other popular devices (logic, ASICs, mPs...), the positive rail operation of ECL was promoted. PECL - Positive Emitter Coupled Logic, also sometime referred to as Pseudo ECL is really just operating the ECL devices between and positive voltage and ground, vs ground and a negative voltage. LVPECL - Low Voltage PECL - is the term used to describe PECL that is powered from a 3.3V power supply. There are even other versions available today that support operation from rails less than 3.3V

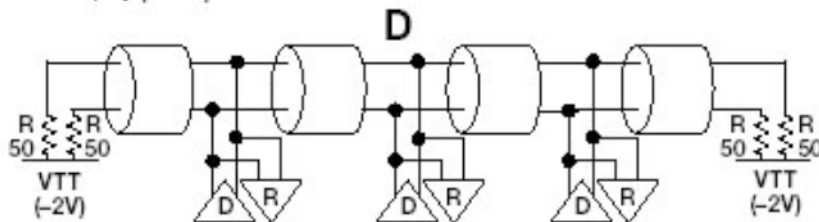
Summary (TI-Bus Interface technical notes)



PECL terminated by Thevenin network, $R_1/R_2 = 50$ Ohms



Differential ECL terminated by 100 Ohm parallel termination



Differential ECL Multipoint bus terminated by 50 Ohms to VTT at each end of the bus.

**LVDS, CML, ECL-
differential interfaces with
odd voltages

BiCMOS

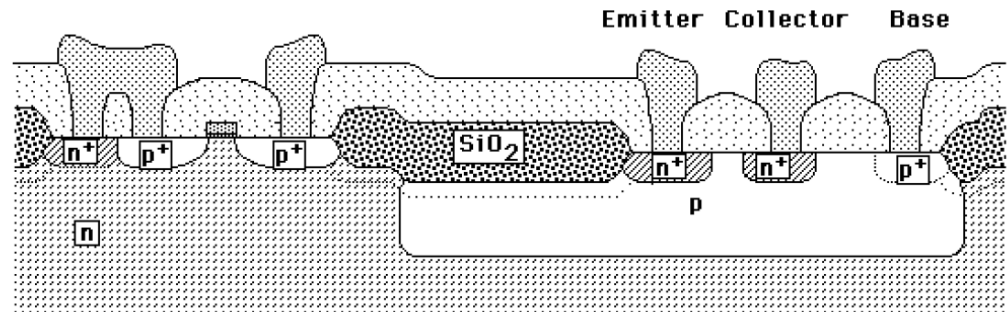
- Concept: MOSFET I_{DS} is amplified by BJT β .
 - CMOS: $I_{out} = I_{DS}$
 - BiCMOS: $I_{out} = \beta * I_{DS}$
- BiCMOS technology combines bipolar and CMOS on one IC chip.
- This technology combines the high-input-impedance, low-power characteristics of CMOS with the high-current drive characteristics of bipolar circuits.
- If the CMOS circuit has to drive a few other similar CMOS logic circuits, the current drive capability is not a problem. However, if a circuit has to drive a relatively large capacitive load, bipolar circuits are preferable because of the relatively large transconductance of BJTs.

BiCMOS

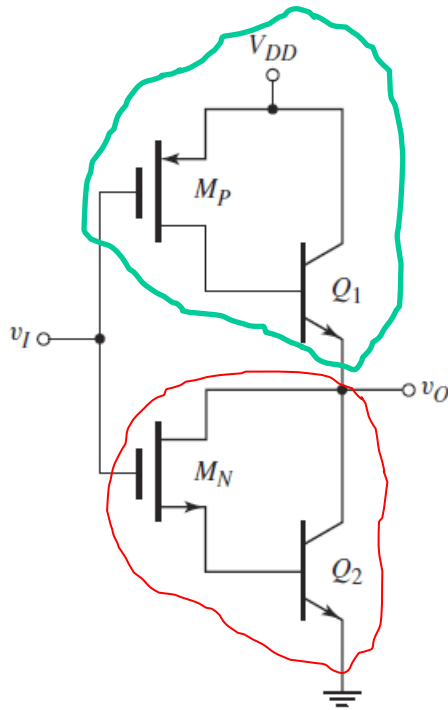
BiCMOS Advantages:

- Improved speed & robustness over CMOS
- Lower power dissipation than bipolar
- Flexible I/O's
- Higher performance analog IC's
- Latchup robustness

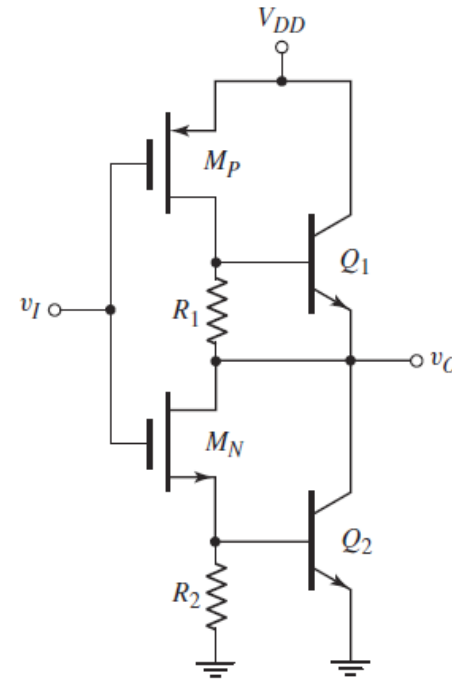
BJT in p-well CMOS process



BiCMOS Inverter

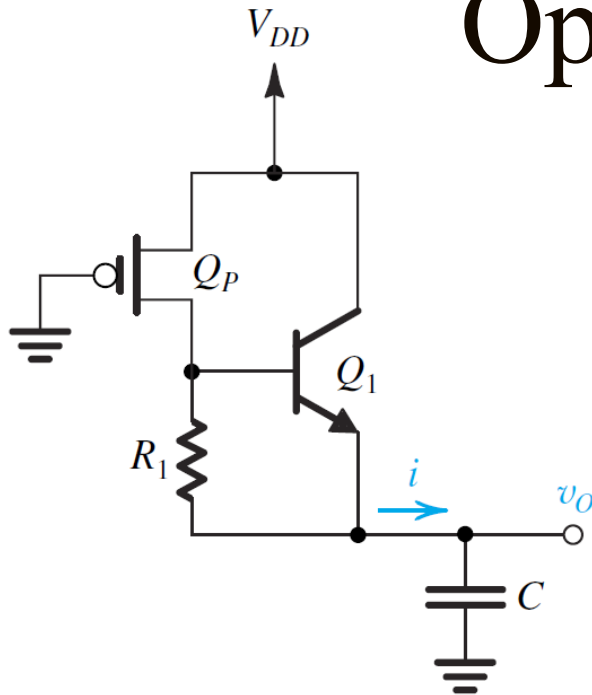


The basic concept is to use an additional bipolar transistor to increase the output current drive of each of M_N and M_P of the CMOS inverter.

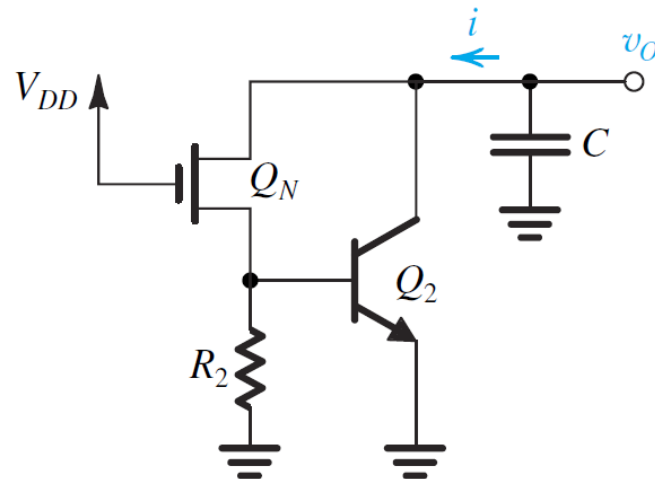


To reduce the turn-off times of M_N and M_P “bleeder resistors” R_1 and R_2 are added. **why???**
Base will float otherwise.

BiCMOS Inverter: Dynamic Operation



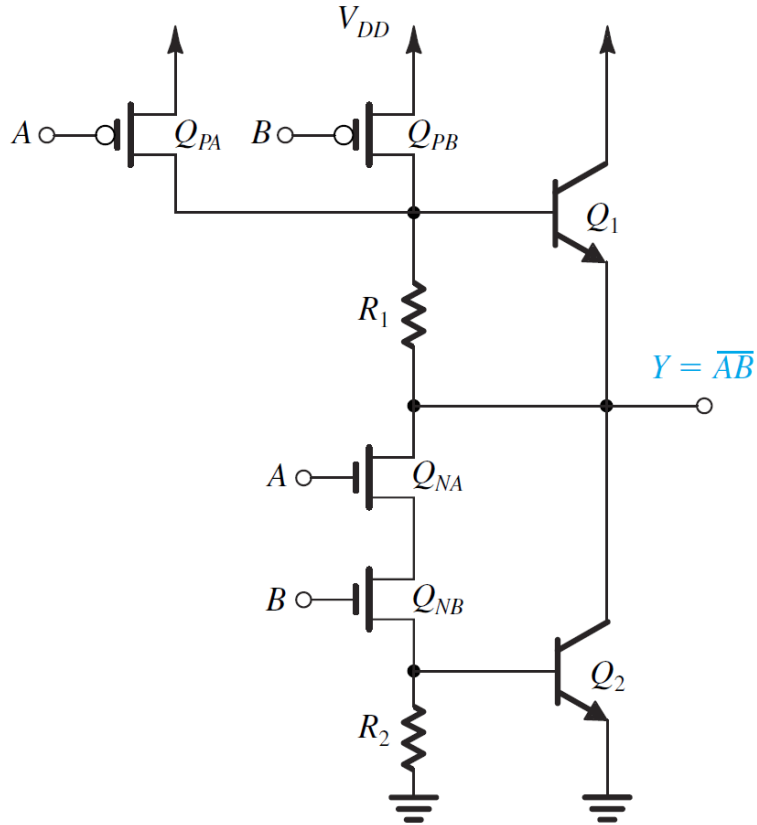
$$V_{OH} = V_{DD} - V_{BE}(FA)$$



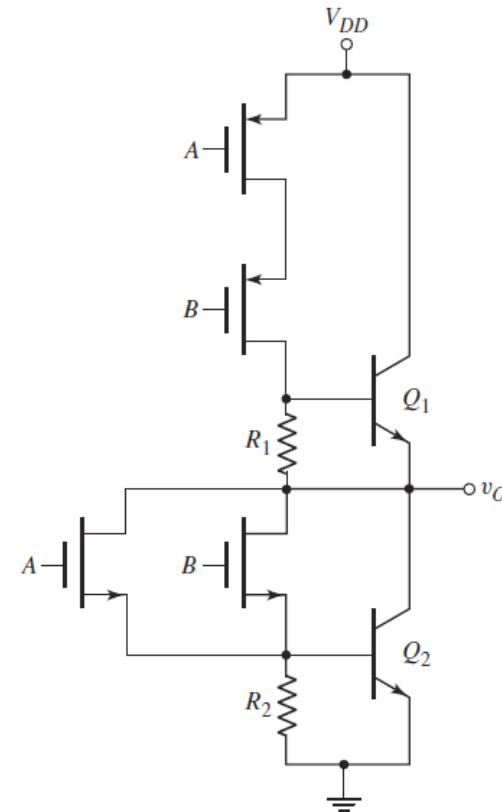
$$V_{OL} = V_{BE}(FA)$$

$V_{BC} \leq 0$, BJTs never saturate

Complex BiCMOS Gates

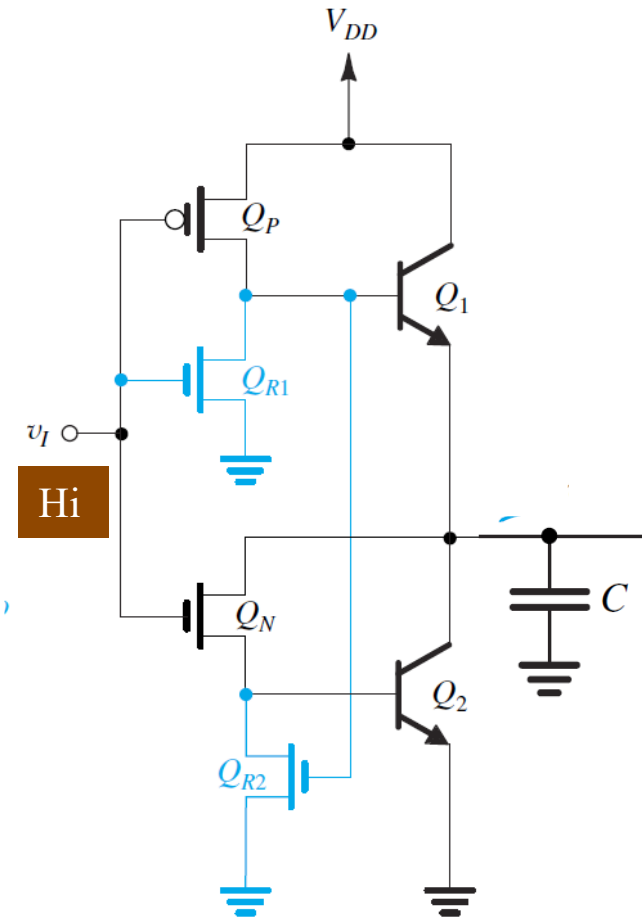
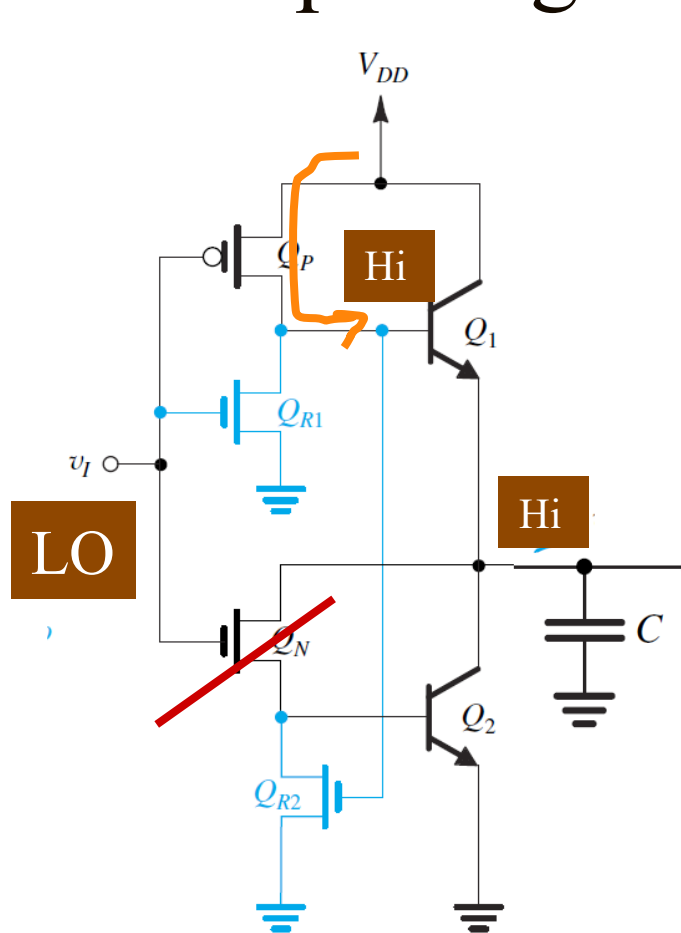


NAND

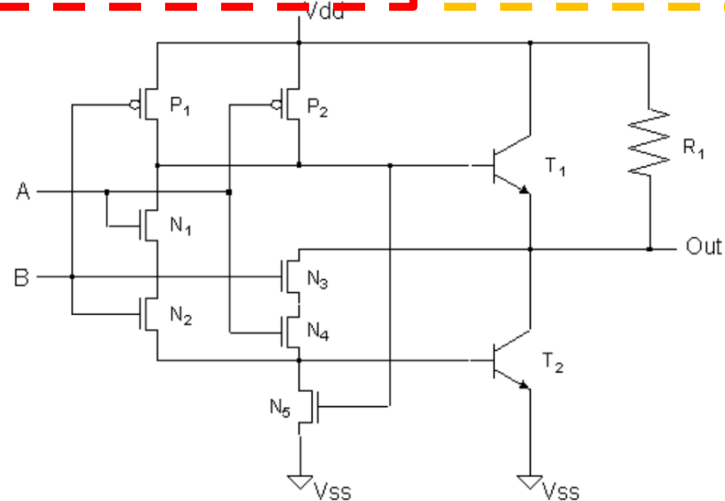


NOR

Replacing Resistors with Transistors



<http://dilvanlab.github.io/old/dilvan/thesis.phd/result.html>



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BiCMOS Issues

- Additional MOSFETs required to control BJT output drivers.
- BJTs are much larger than MOSFETs.
- Lateral BJT has poor β .
- PNP excluded.
 - NPN already requires many additional manufacturing steps.
 - Higher costs, longer production time, lower yield.
 - Results in degraded outputs
- Conclusion: BiCMOS is no longer used in mass production modern processes.
 - Last used $\sim 0.25\mu\text{m}$ technology
- Still used for niche high speed applications.